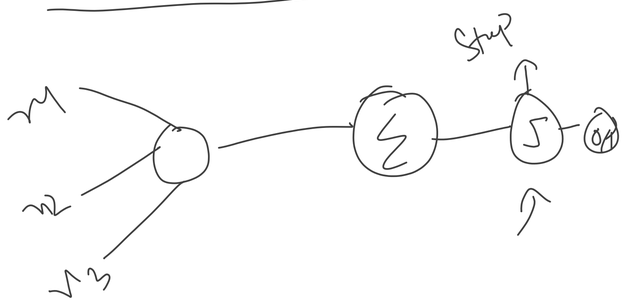
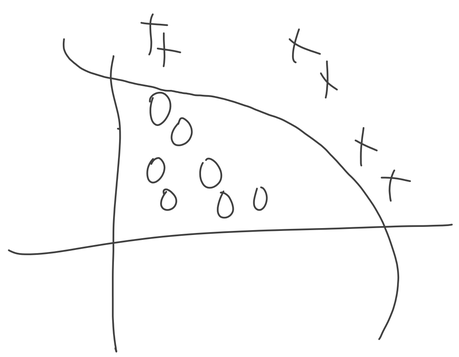
Activation function



$$\begin{cases} 0 & x \geq 0 \\ 1 & x < 0 \end{cases}$$

Activation f^n helps us to add non-linearity



Limitations:

Opt...

L) w_0, w_1, w_2

Sigmoid Activation

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

$x \in (-\infty, \infty)$

$0, 1$

[binary classification]

threshold = 0.5

⇒ Differentiation of Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

[power rule]

$$\frac{d}{dx}\left(\frac{1}{x}\right) = \frac{d}{dx}\left(x^{-1}\right)$$

$$= -x^{-2}$$

$$\frac{d}{dx}\left(\frac{1}{1+e^{-x}}\right)$$

$$= \frac{d}{dx}\left(1+e^{-x}\right)^{-1}$$

$$= -\left(1+e^{-x}\right)^{-2} \cdot \frac{d}{dx}\left(1+e^{-x}\right)$$

$$= \left(\quad \frac{d}{\quad} e^{-x} \right)$$

$$= \frac{-1}{(1+e^{-x})^2} \cdot \left(0 + \frac{d}{dx} e^{-x}\right)$$

$$= -\frac{1}{(1+e^{-x})^2} \cdot \left(e^{-x} \cdot \frac{d}{dx} -x\right)$$

$$= -\frac{1}{(1+e^{-x})^2} e^{-x} \cdot (-1)$$

$$= \frac{e^{x}}{(1+e^{-x})^2}$$

$$= \frac{e^{x}}{(1+e^{-x})(1+e^{-x})}$$

$$= \frac{1}{1+e^{-x}} \cdot \frac{e^{x}}{(1+e^{-x})}$$

$$= \sigma(x) \cdot \frac{e^{x}}{(1+e^{-x})}$$

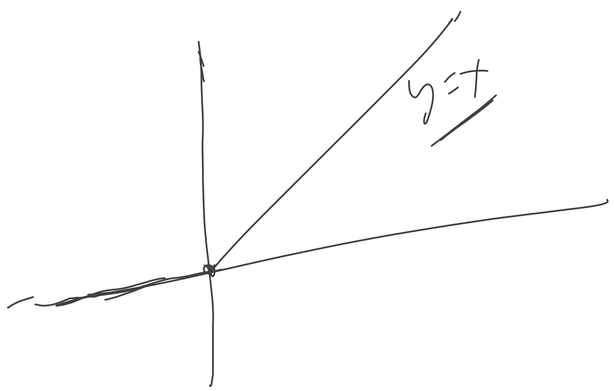
$$= \sigma(x) \cdot \frac{1+e^{-x}-1}{1+e^{-x}}$$

$$= \sigma(x) \left\{ \frac{1+e^{-x}}{1+e^{-x}} - \frac{1}{1+e^{-x}} \right\}$$

$$\left[= \sigma(x)(1 - \sigma(x)) \right]$$

Rectified Linear Unit (ReLU)

$$f(x) = \max(0, x) \quad \begin{cases} 0 & x \le 0 \\ x & x > 0 \end{cases}$$



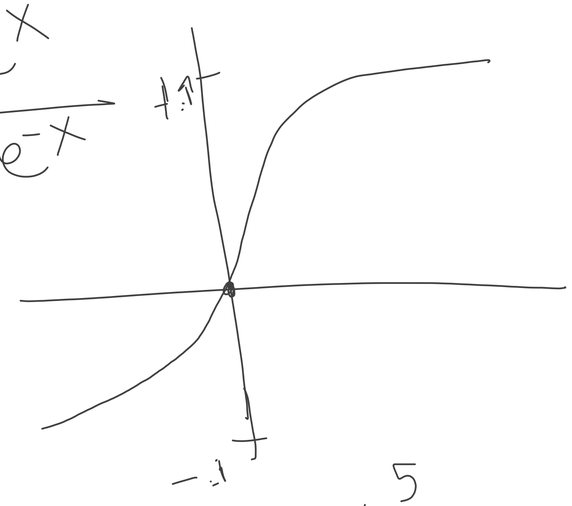
Leaky ReLU: αx

tanh

$-1, 1$

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$\boxed{1 - \tanh^2 x}$$



5

$$\tanh(x) = \frac{\overset{\to u}{e^x - e^{-x}}}{\underset{\to v}{e^x + e^{-x}}}$$

$$h'(x) = \frac{f(x)}{g(x)}$$

$$\left[\frac{u}{v} = \frac{u'v - uv'}{v^2} \right]$$

$$u = e^{x} - e^{-x} \qquad u' = e^{x} + e^{-x}$$

$$v = e^{x} + e^{-x} \qquad v' = e^{x} - e^{-x}$$

$$\frac{d}{dx} e^{x} - e^{-x}$$

$$= e^{x} - (e^{-x}) \cdot (-1)$$

$$= e^{x} + e^{-x}$$

$$\frac{d}{dx} e^{x} + e^{-x}$$

$$= e^{x} + (e^{-x})(-1)$$

$$= e^{x} - e^{-x}$$

$$= \frac{(e^{x}+e^{-x}) \cdot (e^{x}+e^{-x}) - (e^{x}-e^{-x})(e^{x}-e^{-x})}{(e^{x}+e^{-x})^2}$$

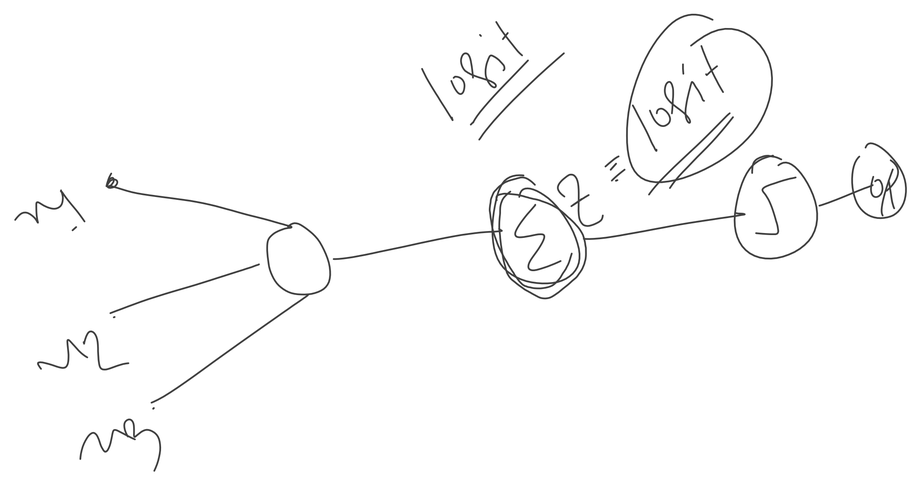
$$= \frac{(e^{x}+e^{-x})^2 - (e^{x}-e^{-x})^2}{(e^{x}+e^{-x})^2}$$

$$= \frac{(e^{x}+e^{-x})^2}{(e^{x}+e^{-x})^2} - \frac{(e^{x}-e^{-x})^2}{(e^{x}+e^{-x})^2}$$

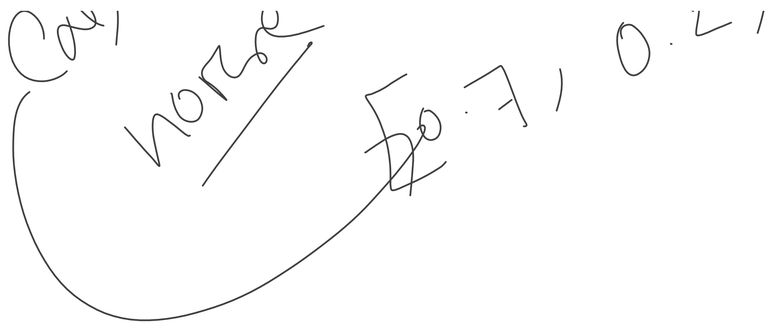
$$= 1 - \left(\frac{e^{x} - e^{-x}}{e^{x} + e^{x}}\right)^2$$

$$= 1 - \tanh^2(x)$$

[Softmax]



[0.1 , 0.8]



Cat
dog
horse

$$\sigma(z_i) = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}$$

$$z = [2, 1, 0]$$

$e^1 = 2.72$ $e^0 = 1$

$e^{\frac{-t}{\cdots}}$

Sum of exponentials:

$7.39 + 2.72 + 1 = 11.11$

$$6(t)_1 = \frac{7.39}{11.11}$$

$$6(t)_2 = \frac{2.72}{11.11}$$

$$6(t)_3 = \frac{1}{11.11}$$